

3. [Maximum mark: 6]

A supermarket analyses the shopping habits of its customers.

The number of times, X , each customer visits the supermarket in a week is given by the following probability distribution.

x	1	2	3	4	5	≥ 6
$P(X = x)$	$1.5a$	$2a$	0.281	a	0.026	0

- (a) (i) Find the value of a . [3]
(ii) Write down the mode of X . [3]
- (b) Find the mean of X . [2]

The manager wants to know why customers come to their supermarket. They survey the first 50 customers to arrive at the supermarket on a particular day.

- (c) Identify which one of the following best describes the manager's sampling method. Circle your answer. [1]

Simple random / Systematic / Convenience / Quota / Stratified

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